

Cryptography Assignment - 4

Decrypt the following ciphertext using Hill Cipher

1. Ciphertext: HLD TQI HTDX WQCMAG

$$\text{Key} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \quad \text{Use } (A=0, B=1, \dots, Z=25)$$

Find the Plaintext

Ans $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \Rightarrow 21-10=11$

 $\Rightarrow 11$ is a coprime with 26

$$\Rightarrow 11^{-1} = 19 \pmod{26}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \pmod{26} \rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$HL = [7, 11] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} = \begin{bmatrix} 21+154 \\ 63+55 \end{bmatrix} = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 4 \end{bmatrix} \begin{matrix} T \\ O \end{matrix}$$

$$DT = [3, 19] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \pmod{26} = \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{matrix} P \\ S \end{matrix}$$

$$GI = [16, 8] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \pmod{26} = \begin{bmatrix} 64 \\ 148 \end{bmatrix} \pmod{26} = \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{matrix} R \\ C \end{matrix}$$

$$HJ = [7, 9] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \pmod{26} = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{matrix} R \\ E \end{matrix}$$

$$DX = [3, 23] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 12 \end{bmatrix} = \begin{matrix} T \\ M \end{matrix}$$

$$WQ = [22, 16] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{matrix} E \\ S \end{matrix}$$

$$CM = [2, 12] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} \pmod{26} = \begin{bmatrix} 18 \\ 0 \end{bmatrix} = \begin{matrix} S \\ A \end{matrix}$$

$$AG = [0, 6] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \pmod{26} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{matrix} G \\ E \end{matrix}$$

Plaintext: TOP SECRET MESSAGE

27 Consider the Hill Cipher Key Matrix

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$$

Cipher Text: RXAODY

Find the plaintext for the above cipher text

Ans Decryption

$$x \mapsto P = K^{-1} \times C \pmod{26}$$

$$\begin{aligned} \det(K) &= (24 \times 14) - (19 \times 5) \\ &= 336 - 95 = 241 = 7 \end{aligned}$$

$\Rightarrow 7$ is a coprime with 26

$$= 7^{-1} = 15$$

$$\Rightarrow K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & 105 \\ 315 & 360 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$RX = \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} \pmod{26} = \begin{bmatrix} 57 \\ 557 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} \Rightarrow \begin{matrix} F \\ L \end{matrix}$$

$$AO = \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} \pmod{26} = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{matrix} O \\ W \end{matrix}$$

$$DY = \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{matrix} F \\ R \end{matrix}$$

Plaintext = FLOWER

3 Confusion v/s Diffusion

Analyze the following classical ciphers with respect to confusion and diffusion. Complete the table and justify your classification based on how each cipher transforms the plaintext

P.T.O

Ciphers	Does it change the character / symbol value?	Does it change the character position?	Confusion Diffusion	Justification
Shift Ciphers	Yes	No	Confusion	Substituted via mathematical shift ($p+k$); character order is unaltered
Multiplication Ciphers	Yes	No	Confusion	Substituted via modular multiplication ($p+k$); character order is unaltered
Transposition Ciphers	No	Yes	Diffusion	character retain original value, but their position are scrambled across the text
Affine Ciphers	Yes	No	Confusion	Combines Multiplication & Shift structure ($(14 \cdot p + 6) \bmod 26$); position remains unchanged

Relationship between Substitution & Confusion

In a substitution cipher, the position remains the same, but the actual character value changes.

$A \rightarrow D$

$\therefore$ The relationship b/w the plaintext characters and ciphertext characters is hidden, hence substitution mainly provides Confusion

Relationship b/w Transposition & Diffusion

In transposition, the character itself doesn't change.

$ABCDE \rightarrow CADEB$

$\therefore$ The characters are moved to different position. This spreads the misinformation of the plaintext throughout the ciphertext. Hence transposition mainly provides diffusion.

4 Transposition

Encryption:-

Encrypt the message "the arrow is pointing north direction" using

- Simple

with key = 4

=> writing row by row into 4 columns, then read column by column

T H E A

R R O W

I S P O

I N T I

N G N O

R T H D

I R E C

T I O N

Cipher Text: T R I I N R I T

U R S N G T R E

E G P T N H E O

A W O I O O C N

- Keyword

with keyword = ATLAS

order them A T L A S

from left to right 1 5 3 2 4

T H E A R

R O W I S

P O I N T

I N G N O

R T H D I

R E C T I

O N X X X

Cipher text = T R P I R R O A I N N D T X E W I G H C X R S T O I I X H C O N T E N

- double

key 1: B R E A K

key 2: L O G

Key 1 = BREAK

B R E A K

2 5 3 1 4

T H E A R

R O W I S

P O I N T

I N G N O

R T H D I

R E C T I

O N X X X

Cipher Text: AINNDTX TRPIRROEWIG4CXRS TOTTXHOONTEN

Key 2 = LOG

2 3 1

A I N

N D T

T R P

I R R

O E W

C R S

T O I

T H O

O N T

E N X

Cipher Text: NTPR ~~W~~HSIOTXANTIOICTII OFIDRREG ROHNN

Decryption:-

⇒ Simple Transposition with Key = 10

Given

Cipher Text

TIDIE MRNME REOGO SANOW RSLTPEEEDOSSSNU

AHTUD BIENP GODAE PEIXD ELNSX

Remove Space

TIDIEMRNME REOGOSANOWRSLTPEEEDOSSSNUAHTUDOTENPGODR

EPEIXDE

ELNSX

Total number of characters = 60

key = 10

No of column = 10

No of rows = $60/10 = 6$

Divide ciphertext into 10 groups, each containing 6 letters

T I D I E M
R N M E R E
O G O S A N
O W R S L T
P E E D O
S S S N U A
H T U D B I
E N P G O D
A E P E T X
D E L N S X

Write these as columns. (or) read column-wise and you get.

Plaintext: TROOPS HEADING WEST NEED MORE SUPPLIES
SEND GENERAL DUBOIS MENTO AID

→ Key word → PLAINCIPHERS

VEY FAX ARA ATX HGD

- PLAIN

53124 - Total = 15

Length = $15/5 = 3$

COL 1 = V E Y	P L A I N
COL 2 = F A X	5 3 1 2 4
COL 3 = A R A	H A V E A
COL 4 = A T X	G R E A T
COL 5 = H G D	D A Y X X

Plain Text: HAVE A GREAT DAY

5 Permutation

Find the inverse permutation for -

1) 43152

$$P = (4, 3, 1, 5, 2)$$

$$1 \rightarrow 3 \text{ in position 3}$$

$$\text{similarly } 2 \rightarrow 5$$

$$3 \rightarrow 2$$

$$4 \rightarrow 1$$

$$5 \rightarrow 4$$

$$\text{Inverse Permutation} = (3, 5, 2, 1, 4)$$

2 7413526

$$P = (7, 4, 1, 3, 5, 2, 6)$$

$$1 \rightarrow 3 \quad 5 \rightarrow 5$$

$$2 \rightarrow 4 \quad 6 \rightarrow 7$$

$$3 \rightarrow 1 \quad 7 \rightarrow 1$$

$$4 \rightarrow 2 \quad P^{-1} = (3, 4, 1, 2, 5, 7, 1)$$

3) 4132

$$P = (4, 1, 3, 2)$$

$$1 \rightarrow 2$$

$$2 \rightarrow 4$$

$$3 \rightarrow 3$$

$$4 \rightarrow 1$$

$$P^{-1} = (2, 4, 3, 1)$$

4) 35142

$$P = (3, 5, 1, 4, 2)$$

$$1 \rightarrow 3$$

$$2 \rightarrow 5$$

$$3 \rightarrow 1$$

$$4 \rightarrow 4 \quad P^{-1} = (3, 5, 1, 4, 2)$$

$$5 \rightarrow 2$$

→ keyword → SECRET

Total → 36

Length → 6

Col 1 = I L E D T N

SECRET

S Z 1 4 3 6

Col 2 = H P T R S O

T H I S I S

Col 3 = I I T M A X

A P L A I N

Col 4 = S A X E R X

N T E X T F

Col 5 = T A N O N I

O R D E M O

Col 6 = S N F O T X

N S T R A T

I O N x x x

Read column by column

Plain Text: THIS IS A PLAIN TEXT FOR DEMONSTRATION